

RISE to accelerate open source RISC-V software creation

RISE, a Swedish research institute, recently declared its plan to speed up the creation of open source RISC-V software, which will further progression in the semiconductor sector and reduce reliance on proprietary software.

RISC-V is an open source instruction set architecture allowing anyone to design their own processors. The RISC-V Foundation, a non-profit organisation, maintains and promotes the RISC-V ISA. Amber Huffman, chair of the RISE Project, stated: “The RISE Project is uniting experts with a joint urgency to escalate the RISC-V software ecosystem preparation with the aid of RISC-V International.”

RISE intends to provide aid to startups, SMEs, and academic establishments in the formation of RISC-V software. The institute will also collaborate with industrial partners to devise a RISC-V software creator platform. Gabriele Columbro, general manager of Linux Foundation Europe, stated that “The RISE Project is devoted to providing RISC-V in open source tools and libraries such as LLVM, GCC, and so on to hasten implementation and time-to-market.”

The RISC-V community has appreciated this effort, with several experts considering it a remarkable step in the acceptance of open source software in the semiconductor industry.



Debian 12 ‘Bookworm’ released after two years of development

Debian 12 ‘Bookworm’ is now available to the public after two years of rigorous development. It’s the successor of Debian 11 ‘Bullseye’ which was released in 2021.



This open source operating system will be supported for the next five years and offers users an updated and reliable experience.

Debian 12 ‘Bookworm’ comes with 11,089 new packages, totalling 64,419 packages. Additionally, 43,254 packages have been updated and

6,296 packages removed as obsolete. The disk usage for the new version is 365GB, with over 1.3 billion lines of code.

The core of ‘Bookworm’ is the Linux kernel 6.1 LTS, offering better hardware support and more convenient access to non-free drivers. It also supports Plasma 5.27.5 as its main desktop environment. Users can choose among different desktop environments, including GNOME, KDE Plasma, LXDE, LXQt, MATE, and Xfce.

Debian 12 ‘Bookworm’ packs quite a punch, unveiling the stylish Emerald artwork from Juliette Taka and introducing the Shiny server. It also brings the command-line tool font for a library of 1,500 DFSG-compatible fonts.

Boasting versatility, the release is suitable for use cases ranging from desktops and netbooks to development, cluster and database servers, plus web and storage applications. Debian 12 ‘Bookworm’ is expected to live up to its reputation of being a highly stable and reliable OS. It is available for download, and users can customise their Debian experience and try different environments.

prpl Foundation releases its LCM project

The prpl Foundation has unveiled its Life Cycle Management (LCM) project, a milestone in the open source sphere, that furnishes service providers with open source tools to deploy and remotely administer apps and services on prplOS and Reference Design Kit – Broadband (RDK-B). This major breakthrough means service providers are now able to launch applications and services quickly, whilst delivering a consistent and positive customer experience.

This open source, platform-independent software, developed as part of prpl’s LCM venture, allows for RDK-B and prplOS portability, by utilising systems similar to the popular ‘App Store’. Service providers can also take advantage of the high-level application programming interface (API) for migrating between RDK-B and prplOS. This permits service providers to opt for the ideal platform for their requirements without becoming ensnared in any one specific vendor or technology.

prpl LCM is the only open source program that works with the modern open industry standards, such as the Open Container Initiative (OCI) and the Broadband Forum TR-157, TR-181 and TR-369 USP. This enables compatibility with various protocols like WebPA and CWMP, and ensures that the CPE (customer premises equipment) of service providers remains updated and secure.

The prpl LCM project has been developed with assistance from a proficient open source community for prpl, comprising chip vendors, systems integrators, service providers, and original equipment manufacturers. This partnership is essential for fulfilling the needs of the whole industry, with the view of bringing the open source app store notion to residential CPEs to life.

According to the prpl Foundation president Dr Leonard Dauphinee, “prpl LCM is a game changer yet remains firmly based on open industry standards from